

Solve

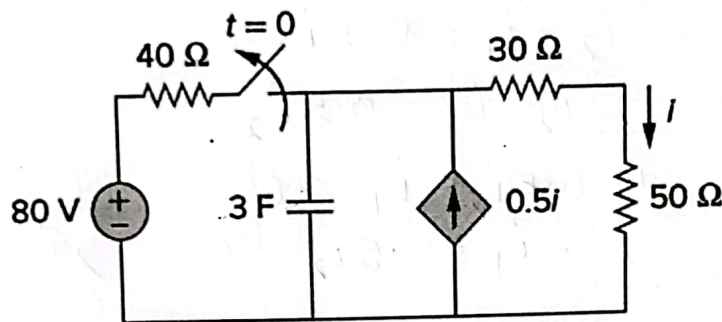
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Brac University
Semester: Spring 2025
Course Code: CSE250
Circuits and Electronics
Section: 07

Assessment: Assignment 4
Full Marks: 20

Question 1 of 2

Consider the circuit shown below.

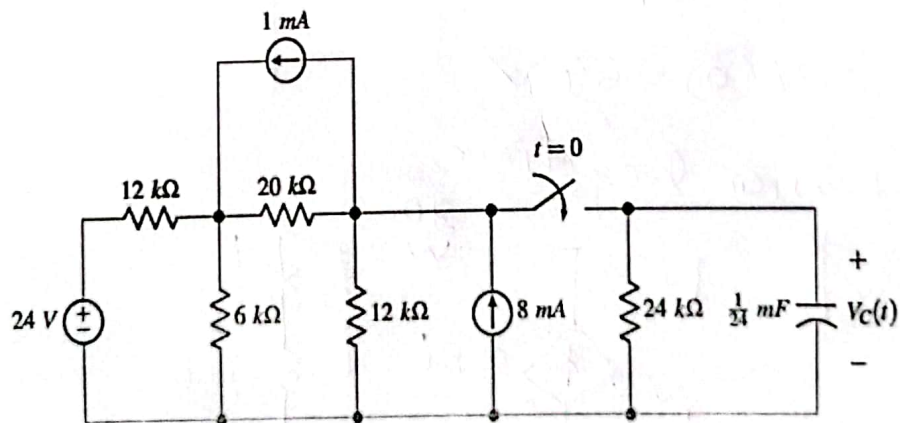


- What will be the $V(0)$ and $V(\infty)$ of the capacitor.
- Find $i(t)$ for $t < 0$ and $t > 0$.

[3]
[2]

Question 2 of 2:

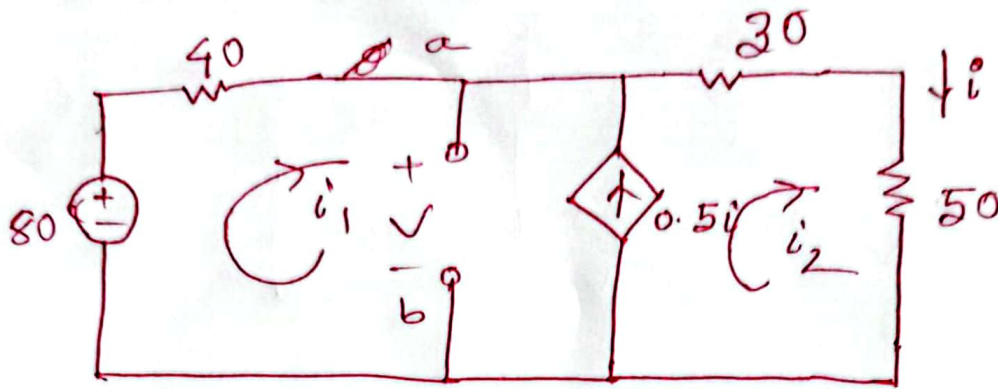
Consider the circuit shown below.



- Find the value of τ .
- Find the equation of $V_c(t)$ and draw the $V_c(t)$ vs t plot.

[2]
[3]

1. a) $t < 0$,



Here, $i_2 = i$, $i_2 - i_1 = 0.5i$

$$\Rightarrow i_2 - i_1 = 0.5 i_2$$

$$\neq 0.5\dot{v}_2 - \dot{v}_1 = 0$$

$$\Rightarrow -i_1 + 0.5i_2 = 0 \rightarrow \textcircled{1}$$

Super mesh,

$$-80 + 40i_1 + 30i_2 + 50i_2 = 0$$

$$\Rightarrow 40i_1 + 80i_2 = 80 \quad \text{---} \textcircled{w}$$

$$i_1 = 0.4, \quad i_2 = 0.8$$

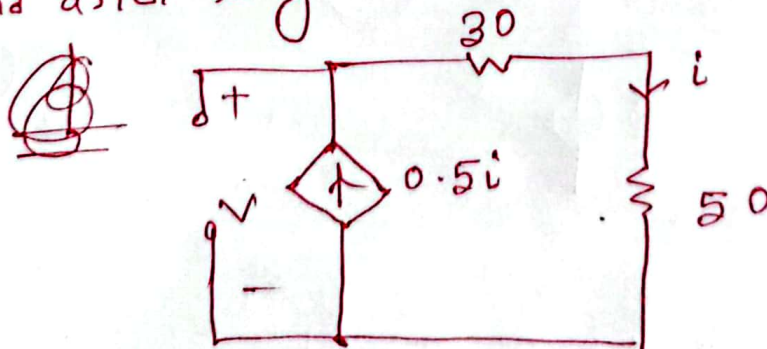
KVL,

$$-80 + 40i_1 + v = 0$$

$$v = 64 \text{ V}$$

$$\therefore V(0) = 64 \checkmark$$

$t > 0$ and after long time,



Here, $i = 0.5i$

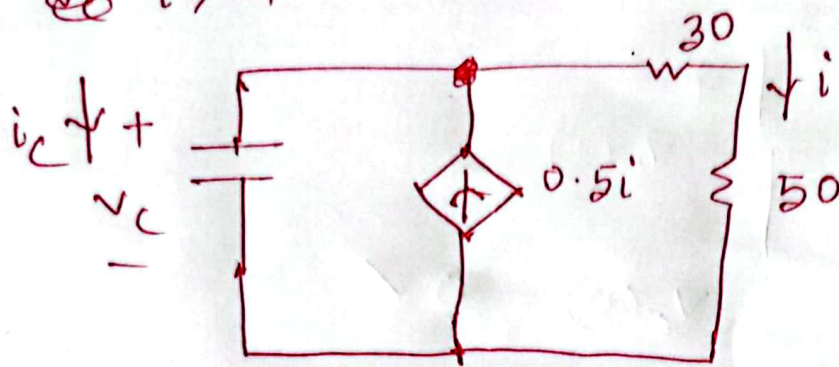
$$0.5i = 0$$

$$\Rightarrow i = 0$$

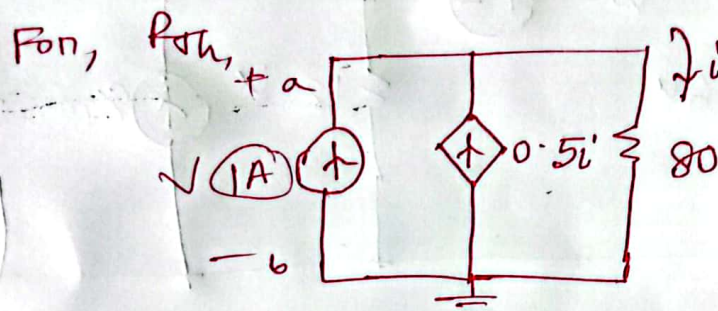
$$- \sqrt{0} + 30 \times 0 + 50 \times 0 = 0$$

$$V = 0, \quad V(\infty) = 0$$

b) $t > 0$,



$$v_c(t) = v(\infty) + [v(0) - v(\infty)] e^{-t/\tau}$$



$$-1 - 0.5i + i = 0$$

$$0.5i = 1$$

$$\Rightarrow i = 2$$

$$V = 80 \times 2 = 160$$

$$R_{th} = \frac{V}{I} = 160 \Omega$$

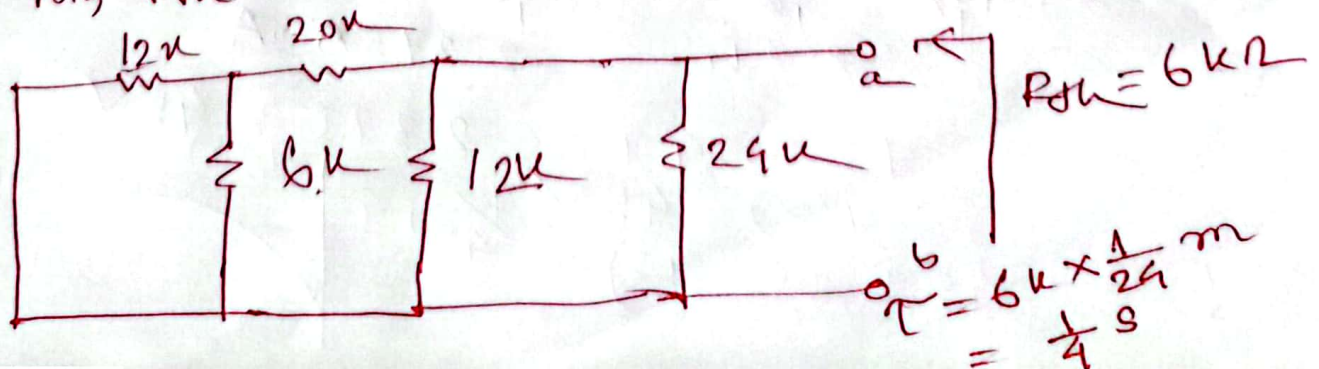
$$\tau = 2 \times 160 = 480 \text{ s}$$

$$v_c(t) = 0 + 64 e^{-t/480}$$

$$i = \frac{v_c(t)}{80} = 0.8 e^{-t/480}, \quad t > 0$$

$$i = i_2 = 0.8 \text{ A} \quad t < 0$$

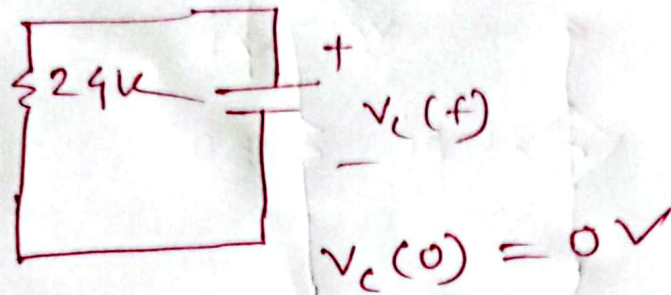
2. a) For, R_{th}



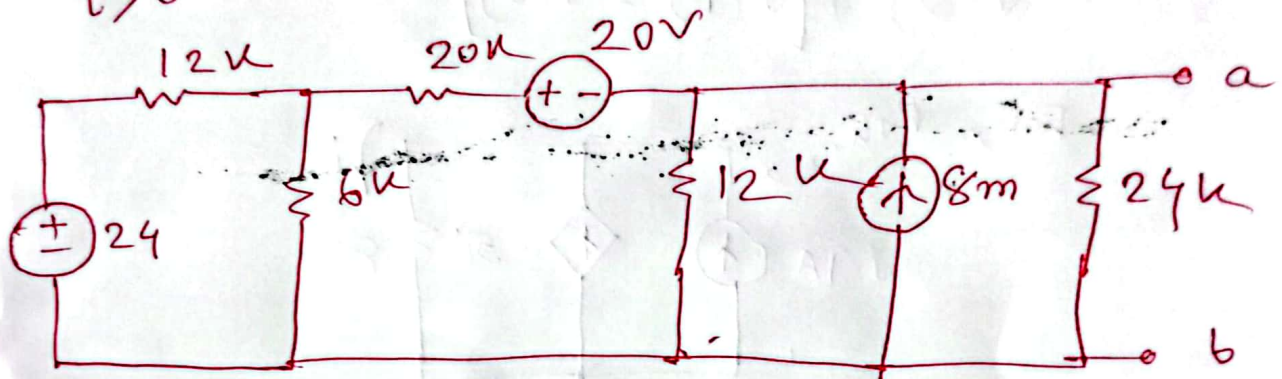
$$R_{th} = 6 \text{ k}\Omega$$

$$\tau = 6 \times \frac{1}{24} \text{ m} = \frac{1}{4} \text{ s}$$

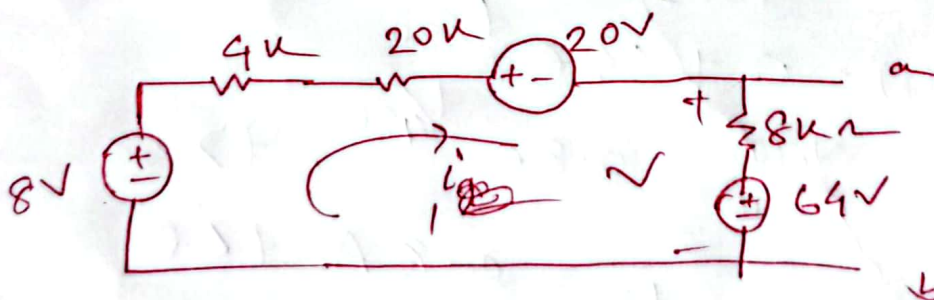
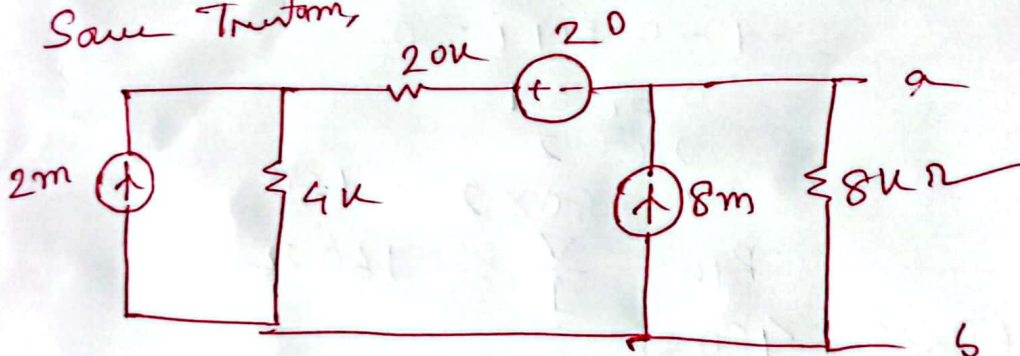
2b) $t < 0$



$t > 0$



Source Transform



$$-8 + 4i_1 + 20i_1 + 20 + 8i_1 + 64 = 0$$

$$i_1 = -2.375 \text{ mA}$$

$$V = +64 + 8 \times -2.375 = 45 \text{ V}$$

$$\therefore V(\infty) = 45 \text{ V}$$

$$V_c(t) = 45 + (0 - 45)e^{-t/480}$$

